

Now, $z_1 = \frac{(4, 2)}{3-i} = \frac{4+2i}{3-i}$

Multiply the numerator and denominator by the complex conjugate of $z_1 = 3-i$.

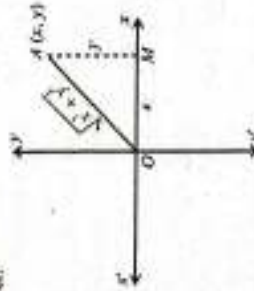
$$z_1 = \frac{4+2i}{3-i} \times \frac{3+i}{3+i} = \frac{(4+2i)(3+i)}{(3-i)(3+i)}$$

$$= \frac{12+4i+6i+2i^2}{9-(-1)} = \frac{12+10i-2}{10} = \frac{10+10i}{10} = 1+i$$

Thus, $z_1 = 1+i$

Modulus of Complex Number: The real number $\sqrt{x^2 + y^2}$ is called the modulus of the complex number $x+iy$ and it is denoted by $|x+iy|$. In Figure, $|OA|$ represents the modulus of $x+iy$. OR In other words, the modulus of a complex number is the distance from the origin to the point representing the number.

In other words, the modulus of a complex number is the distance from the origin to the point representing the number.



Example 3: If $z = \frac{1+2i}{2-i}$ then evaluate $|\bar{z}|$

Sol. $z = \frac{1+2i}{2-i} = \frac{1+4i+4i^2}{2-i} = \frac{-3+4i}{2-i} = \frac{-3+4i}{2-i} \times \frac{2+i}{2+i} = \frac{-6-3i+8i+4i^2}{2^2-i^2} = \frac{-6+5i-4}{4-(-1)} = \frac{-10+5i}{5} = -2+i$

Taking conjugate $\bar{z} = -2-i$
and $|\bar{z}| = |-2-i| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{4+1} = \sqrt{5}$

Note:

- Every real number is a complex number with 0 as its imaginary part.
- A real number is self-conjugate.
- The set C of complex numbers does not satisfy the order axioms. In fact, there is no sense in saying that one complex number is greater or less than the other.

Exercise 1.1

Q1 Find the multiplicative inverse of each of the following numbers:

(i) $(-4, 7)$

Sol. Given that $z = (-4, 7)$

$\Rightarrow z = -4+7i$

So, $\frac{1}{z} = \frac{1}{-4+7i} \Rightarrow \frac{1}{z} = \frac{1}{-4+7i} \times \frac{-4-7i}{-4-7i} = \frac{-4-7i}{16-49i^2} = \frac{-4-7i}{16+49} = \frac{-4-7i}{65} = -\frac{4}{65} - \frac{7}{65}i$

(ii) $(\sqrt{2}, -\sqrt{5})$

Sol. Given that $z = (\sqrt{2}, -\sqrt{5})$

$\Rightarrow z = \sqrt{2} - \sqrt{5}i$

So, $\frac{1}{z} = \frac{1}{\sqrt{2} - \sqrt{5}i} \Rightarrow \frac{1}{z} = \frac{1}{\sqrt{2} - \sqrt{5}i} \times \frac{\sqrt{2} + \sqrt{5}i}{\sqrt{2} + \sqrt{5}i} = \frac{\sqrt{2} + \sqrt{5}i}{2 - 5i^2} = \frac{\sqrt{2} + \sqrt{5}i}{2+5} = \frac{\sqrt{2} + \sqrt{5}i}{7} = \frac{\sqrt{2}}{7} + \frac{\sqrt{5}}{7}i$

(iii) $(1, 0)$

Sol. Given that $z = (1, 0)$

$\Rightarrow z = 1+0i$

So, $\frac{1}{z} = 1 \Rightarrow \frac{1}{z} = (1, 0)$

Q2 Separate into real and imaginary parts (write as a simple complex number):

(i) $\frac{2-7i}{4+5i}$

Sol. Let $z = \frac{2-7i}{4+5i}$

$= \frac{2-7i}{4+5i} \times \frac{4-5i}{4-5i} = \frac{2(4-5i) - 7i(4-5i)}{(4+5i)(4-5i)}$

$= \frac{8-10i-28i+35i^2}{16-25i^2} = \frac{8-10i-28i+35(-1)}{16-25(-1)} = \frac{8-10i-28i-35}{16+25}$

$= \frac{-27-38i}{41} = -\frac{27}{41} - \frac{38}{41}i$

So, $\text{Re}(z) = -\frac{27}{41}$ and $\text{Im}(z) = -\frac{38}{41}$

(ii) $\frac{(-2+3i)^2}{1+i}$

Sol. Let $z = \frac{(-2+3i)^2}{1+i}$

$= \frac{(-2)^2 + (3i)^2 + 2(-2)(3i)}{1+i} = \frac{4+9i^2-12i}{1+i} = \frac{4-9-12i}{1+i}$

$= \frac{-5-12i}{1+i} = \frac{-5-12i}{1+i} \times \frac{1-i}{1-i} = \frac{-5(1-i)-12i(1-i)}{1^2-i^2} = \frac{-5+5i-12i+12i^2}{1-(-1)}$

$= \frac{-5+5i-12i+12(-1)}{1+1} = \frac{-5+5i-12i-12}{2} = \frac{-17-7i}{2} = -\frac{17}{2} - \frac{7}{2}i$

So, $\text{Re}(z) = -\frac{17}{2}$ and $\text{Im}(z) = -\frac{7}{2}$

(iii) $\frac{i}{1+i}$

Sol. Let $z = \frac{i}{1+i}$

$= \frac{i}{1+i} \times \frac{1-i}{1-i} = \frac{i(1-i)}{1^2-i^2} = \frac{i-i^2}{1-(-1)} = \frac{i+1}{2} = \frac{1}{2} + \frac{i}{2}$

So, $\text{Re}(z) = \frac{1}{2}$ and $\text{Im}(z) = \frac{1}{2}$

(iv) $\frac{(4+3i)^2}{4-3i}$

Sol. Let $z = \frac{(4+3i)^2}{4-3i}$

$= \frac{(4)^2 + (3i)^2 + 2(4)(3i)}{4-3i} = \frac{16+9i^2+24i}{4-3i} = \frac{16-9+24i}{4-3i} = \frac{7+24i}{4-3i}$

$= \frac{7+24i}{4-3i} \times \frac{4+3i}{4+3i} = \frac{7(4+3i) + 24i(4+3i)}{(4-3i)(4+3i)} = \frac{28+21i+96i+72i^2}{16-9i^2}$

$= \frac{28+21i+96i+72(-1)}{16-9(-1)} = \frac{28+21i+96i-72}{16+9}$

$= \frac{-44+117i}{25} = -\frac{44}{25} + \frac{117}{25}i$

So, $\text{Re}(z) = -\frac{44}{25}$ and $\text{Im}(z) = \frac{117}{25}$

Q3 Prove that $\bar{z} = z$ iff z is real.

Sol. Assume that $z = a+bi$

then $\bar{z} = a-bi$

Given $\bar{z} = z$

$a-bi = a+bi$

$a-bi-a-bi=0$

$-2bi=0 \Rightarrow b=0$

So, $z = a+0i = a$

Hence z is a real number if $\bar{z} = z$.

Conversely,

Assume that z is a real number.

$z = a \Rightarrow \bar{z} = a = z$

Hence $\bar{z} = z$ if z is a real number.

Q4 For $z \in C$, show that:

(i) $\frac{z+\bar{z}}{2} = \text{Re}(z)$

Sol. Take, L.H.S. $= \frac{z+\bar{z}}{2}$

Assume that $z = a+bi$

So, L.H.S. $= \frac{(a+bi) + (a-bi)}{2} = \frac{a+bi+a-bi}{2} = \frac{2a}{2} = a$

L.H.S. $= \text{Re}(z)$

L.H.S. $= \text{R.H.S.}$

(iv) $\frac{z-2}{2i} = \text{Im}(z)$

Sol. L.H.S. = $\frac{z-2}{2i}$

A.S. $z = a + bi$

So, L.H.S. = $\frac{(a+bi)-2}{2i}$

= $\frac{(a-2)+bi}{2i}$

= $\frac{a-2}{2i} + \frac{bi}{2i}$

= $\frac{a-2}{2i} + \frac{b}{2}$

L.H.S. = R.H.S.

(iii) $|z|^2 = z \cdot \bar{z}$

Sol. Assume that $z = a + bi$

L.H.S. = $|z|^2$

= $(a+bi)(a-bi)$

= $a^2 - (bi)^2$

= $a^2 - b^2(-1)$

= $a^2 + b^2$

From (i) and (ii)

L.H.S. = R.H.S.

Q5. If $z_1 = 2 + i$, $z_2 = 3 - 2i$, $z_3 = 1 + 3i$ then express

$\frac{z_1 z_2}{z_3}$ in the form of $a + bi$.

Sol. A.S. $z_1 = 2 + i$, $z_2 = 3 - 2i$, $z_3 = 1 + 3i$

$\frac{z_1 z_2}{z_3} = \frac{(2+i)(3-2i)}{1+3i}$

= $\frac{(6-4i+3i-2i^2)}{1+3i}$

= $\frac{(6-4i+3i-2(-1))}{1+3i}$

= $\frac{(6-4i+3i+2)}{1+3i}$

= $\frac{(8-i)}{1+3i}$

= $\frac{(8-i)(1-3i)}{(1+3i)(1-3i)}$

= $\frac{(8-24i-1+3i)}{1-9i^2}$

= $\frac{(7-21i)}{1+9}$

= $\frac{(7-21i)}{10}$

= $\frac{7}{10} - \frac{21}{10}i$

= $\frac{7}{10} - 2.1i$

= $\frac{7}{10} - 2.1i$

= $\frac{7}{10} - 2.1i$

= $\frac{7}{10} - 2.1i$

= $\frac{7}{10} - 2.1i$

= $\frac{7}{10} - 2.1i$

Q6. If $z_1 = 2 + 7i$ and $z_2 = -5 + 3i$, then evaluate the following:

(i) $12z_1 - 4z_2$

Sol. $12z_1 - 4z_2 = 12(2+7i) - 4(-5+3i)$

= $24 + 84i + 20 - 12i$

= $44 + 72i$

(ii) $13z_1 + 2z_2$

Sol. $13z_1 + 2z_2 = 13(2+7i) + 2(-5+3i)$

= $26 + 91i - 10 + 6i$

= $16 + 97i$

(iii) $7z_1 + 4z_2$

Sol. $7z_1 + 4z_2 = 7(2+7i) + 4(-5+3i)$

= $14 + 49i - 20 + 12i$

= $-6 + 61i$

(iv) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(v) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(vi) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(vii) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(viii) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(ix) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

(x) $10z_1 + 2z_2$

Sol. $10z_1 + 2z_2 = 10(2+7i) + 2(-5+3i)$

= $20 + 70i - 10 + 6i$

= $10 + 76i$

Q7. Find the least positive value of n , if

$\left(\frac{1+i}{1-i}\right)^n = 1$

Sol. $\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

$\left(\frac{1+i}{1-i}\right)^n = 1$

Comparing real and imaginary part, we have

$2x - 2y = 5 \dots (i)$

$2x + 3y = 12 \dots (ii)$

Multiplying equation (i) by 3 and equation (ii) by 2, we have

$6x - 6y = 15$

$4x + 6y = 24$

Add the equations

$9x - 6y + 4x + 6y = 15 + 24$

$13x = 39$

$x = 3$

Substitute $x = 3$ in equation (i), we have

$3(3) - 2y = 5$

$9 - 2y = 5$

$-2y = -4$

$y = 2$

Thus, $x = 3, y = 2$

Example 2: Find the square root of complex number $5 + 12i$, and also represent the square roots on an Argand diagram.

Sol. Let $x + yi = 5 + 12i$

$\Rightarrow x = 5$ and $y = 12 > 0$

$|z| = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = 13$

Applying the square root formula for complex numbers, we get

$\sqrt{5 + 12i} = \pm \left(\sqrt{\frac{13+5}{2}} + i \sqrt{\frac{13-5}{2}} \right)$

$= \pm \left(\sqrt{9} + i \sqrt{4} \right) = \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

$= \pm (3 + 2i)$

Exercise 1.2

Q1. Find the values of x and y in each of the following

(i) $x + yi = 2 - 3i$

Sol. $x + yi = 2 - 3i$

$\Rightarrow x = 2, y = -3$

$\Rightarrow x = 2, y = -3$

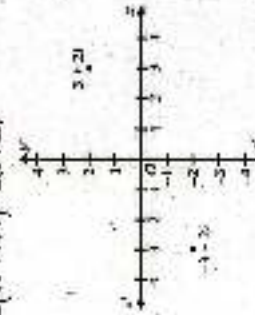
$\Rightarrow x = 2, y = -3$

$\Rightarrow x = 2, y = -3$

$\Rightarrow x = 2, y = -3$

$\Rightarrow x = 2, y = -3$

Thus, the square root of the complex number $5 + 12i$ are $3 + 2i$ and $-3 - 2i$ are shown in adjacent figure.



1.2 Equality of Two Complex Numbers

Example 1: If $(3 + 2i)(x + yi) = 8 + 12i$, where $x, y \in \mathbb{R}$, then find the values of x and y . Complex Numbers

Sol. Given that $(3 + 2i)(x + yi) = 8 + 12i$

$\Rightarrow 3x + 3iy + 2ix + 2iy^2 = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

$\Rightarrow (3x - 2y) + (3y + 2x)i = 8 + 12i$

Hence, proved

Q2. Show that, the value of r for $n \in \mathbb{N}$ and $n > 4$ is r , where r is the remainder when n is divided by 4.

Sol. Proof:

A.S. given $n \in \mathbb{N}$ and $n > 4$. Also, r is the remainder when n is divided by 4.

So, we can write that $n = 4k + r$ where $n \in \mathbb{N}$.

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$r = n - 4k$

$$= \pm \sqrt{\frac{\frac{130}{2} \pm \sqrt{\frac{130^2}{4} - \frac{100 \times 10}{2}}}{2}}$$

$$= \pm \sqrt{\frac{\frac{130}{2} \pm \sqrt{\frac{130^2}{4} - \frac{100 \times 10}{2}}}{2}}$$

$$= \pm \sqrt{\frac{\frac{130}{2} \pm \sqrt{\frac{130^2}{4} - \frac{100 \times 10}{2}}}{2}}$$

$$= \pm \sqrt{\frac{\frac{130}{2} \pm \sqrt{\frac{130^2}{4} - \frac{100 \times 10}{2}}}{2}}$$

Either

$$x + iy = \sqrt{\frac{130 - 7}{20}} + i \sqrt{\frac{130 + 7}{20}}$$

$$\text{or } x + iy = \sqrt{\frac{130 - 7}{20}} + i \sqrt{\frac{130 + 7}{20}}$$

By comparing the real and imaginary parts

$$x = \sqrt{\frac{130 - 7}{20}} \quad \text{and} \quad y = \sqrt{\frac{130 + 7}{20}}$$

$$\text{or } x = \sqrt{\frac{130 - 7}{20}} \quad \text{and} \quad y = \sqrt{\frac{130 + 7}{20}}$$

10. If $x_1 = 2 + 3i$ and $x_2 = 1 - \alpha$, find the value of α such that $\text{Im}(x_1 x_2) = 7$.

Sol. For $x_1 x_2$

$$x_1 x_2 = (2 + 3i)(1 - \alpha)$$

$$x_1 x_2 = 2 - 2\alpha + 3i - 3\alpha i$$

$$x_1 x_2 = (2 - 2\alpha) + (3 - 3\alpha)i$$

$$\text{Re}(x_1 x_2) = 2 - 2\alpha \quad \text{and} \quad \text{Im}(x_1 x_2) = 3 - 3\alpha$$

$$\text{Given } \text{Im}(x_1 x_2) = 7$$

Put the value of $\text{Im}(x_1 x_2)$

$$3 - 3\alpha = 7$$

$$-3\alpha = 7 - 3$$

$$-3\alpha = 4$$

$$\alpha = -\frac{4}{3}$$

$$\alpha = -\frac{4}{3}$$

10. If $x_1 = x + yi$ and $x_2 = x + yi$, find x, y and b such that $x_1 + x_2 = 10 + 4i$ and $x_1 - x_2 = 6 + 2i$.

Sol. Given that $x_1 - x_2 = 10 + 4i$

$$x + yi - (x + yi) = 10 + 4i$$

$$x + yi - x - yi = 10 + 4i$$

By comparing the real and imaginary parts

$$x - x = 10 \quad \text{--- (i)}$$

$$y - y = 4 \quad \text{--- (ii)}$$

Also

$$x_1 - x_2 = 6 + 2i$$

$$x - yi - (x + yi) = 6 + 2i$$

$$x - yi - x - yi = 6 + 2i$$

$$x - x - yi - yi = 6 + 2i$$

$$(x - x) + (-y - y)i = 6 + 2i$$

$$x - x = 6 \quad \text{--- (i)}$$

$$y - y = 2 \quad \text{--- (ii)}$$

Add (i) and (ii)

$$x + x + x - x = 10 + 6$$

$$2x - 10 = 16 \Rightarrow x = \frac{16}{2} \Rightarrow x = 8$$

Put in (i)

$$x + x = 10$$

$$8 + x = 10$$

$$x = 10 - 8$$

$$x = 2$$

Add (i) and (ii)

$$y - y + y - y = -4 + 2$$

$$2y = 6 \Rightarrow y = \frac{6}{2} \Rightarrow y = 3$$

Put in (ii)

$$y - y = 4$$

$$3 - y = 4$$

$$y = 3 - 4$$

$$y = -1$$

10. Show that $\forall x_1, x_2 \in \mathbb{C}$, $\frac{x_1}{x_2} = \frac{\bar{x}_1}{\bar{x}_2}$

Sol. Assume that $x_1 = a + bi$ and $x_2 = c + di$

L.H.S. $= \frac{x_1}{x_2}$

$$= \frac{(a + bi)(a - bi)}{(c + di)(c - di)}$$

$$= \frac{a^2 - b^2 + 2abi}{c^2 - d^2 + 2cdi}$$

$$= \frac{a^2 - b^2 + 2abi}{c^2 - d^2 + 2cdi} \cdot \frac{c^2 - d^2 - 2cdi}{c^2 - d^2 - 2cdi}$$

$$= \frac{(a^2 - b^2 + 2abi)(c^2 - d^2 - 2cdi)}{(c^2 - d^2)^2 - (2cd)^2}$$

$$= \frac{a^2 c^2 - a^2 d^2 - 2abcdi + 2abcdi - b^2 c^2 + b^2 d^2}{c^4 - d^4 - 4c^2 d^2}$$

$$= \frac{a^2 c^2 - a^2 d^2 - b^2 c^2 + b^2 d^2}{c^4 - d^4 - 4c^2 d^2}$$

$$= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$$

$$= (a_1 a_2 - b_1 b_2) - (a_1 b_2 + a_2 b_1)i$$

R.H.S. $= z_1 z_2$

$$= (a_1 + bi)(a_2 + bi)$$

$$= a_1 a_2 - a_1 b_2 i - a_2 b_1 i + b_1 b_2 i^2$$

$$= a_1 a_2 - a_1 b_2 i - a_2 b_1 i + b_1 b_2 (-1)$$

$$= a_1 a_2 - a_1 b_2 i - a_2 b_1 i - b_1 b_2$$

$$= a_1 a_2 - a_1 b_2 i - a_2 b_1 i - b_1 b_2$$

$$= a_1 a_2 - a_1 b_2 i - a_2 b_1 i - b_1 b_2$$

$$= (a_1 a_2 - b_1 b_2) - (a_1 b_2 + a_2 b_1)i$$

From (i) and (ii)

L.H.S. = R.H.S.

11. Find the square root of the following complex numbers:

$$(i) -7 - 24i$$

Sol. By using the square root formula for complex numbers

$$\sqrt{x + yi} = \sqrt{\frac{x + \sqrt{x^2 + y^2}}{2}} + i \sqrt{\frac{-x + \sqrt{x^2 + y^2}}{2}}$$

$$\sqrt{-7 - 24i}$$

$$= \sqrt{\frac{-7 + \sqrt{(-7)^2 + (-24)^2}}{2}} + i \sqrt{\frac{-(-7) + \sqrt{(-7)^2 + (-24)^2}}{2}}$$

$$= \sqrt{\frac{-7 + \sqrt{49 + 576}}{2}} + i \sqrt{\frac{7 + \sqrt{49 + 576}}{2}}$$

$$= \sqrt{\frac{-7 + \sqrt{625}}{2}} + i \sqrt{\frac{7 + \sqrt{625}}{2}}$$

$$= \sqrt{\frac{-7 + 25}{2}} + i \sqrt{\frac{7 + 25}{2}}$$

$$= \sqrt{\frac{18}{2}} + i \sqrt{\frac{32}{2}}$$

$$= \sqrt{9} + i \sqrt{16}$$

$$= 3 + 4i$$

Either

$$\sqrt{-7 - 24i} = \sqrt{\frac{-7 + \sqrt{49 + 576}}{2}} + i \sqrt{\frac{-(-7) + \sqrt{49 + 576}}{2}}$$

Sol. By using the square root formula for complex numbers

$$\sqrt{x + yi} = \sqrt{\frac{x + \sqrt{x^2 + y^2}}{2}} + i \sqrt{\frac{-x + \sqrt{x^2 + y^2}}{2}}$$

Either

$$\sqrt{-15-36i} = \pm(\sqrt{3-3\sqrt{3}}) + i\sqrt{-15-36i} = \pm(2\sqrt{3}-3\sqrt{3})$$

$$\sqrt{-15-36i} = 2\sqrt{3}-3\sqrt{3} \text{ or } \sqrt{-15-36i} = -2\sqrt{3}+3\sqrt{3}$$

(iv) $119 + 120i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{(119)^2+(120)^2}{2}} + \frac{i(120)}{|120|} \sqrt{\frac{(119)^2+(120)^2}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{14161+14400}{2}} + \frac{i(120)}{120} \sqrt{\frac{14161+14400}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{28561}{2}} + i\sqrt{\frac{28561}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{169+119}{2}} + i\sqrt{\frac{169-119}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{288}{2}} + i\sqrt{\frac{50}{2}} \right)$$

$$\sqrt{119+120i} = \pm (\sqrt{144} + i\sqrt{25})$$

$$\sqrt{119+120i} = \pm (12+5i)$$

Either

$$\sqrt{119+120i} = \pm (12+5i) \text{ or } \sqrt{119+120i} = \pm (12-5i)$$

Q3. Find the square root of $13-20\sqrt{3}i$ and represent them on an Argand diagram.

Sol. By using the square root formula for complex numbers

$$\sqrt{x+iy} = \pm \left(\sqrt{\frac{x^2+y^2}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{(13)^2+(20\sqrt{3})^2}{2}} + \frac{i(-20\sqrt{3})}{|20\sqrt{3}|} \sqrt{\frac{(13)^2+(20\sqrt{3})^2}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm$$

$$\sqrt{\frac{\sqrt{169+1200+13} - 20\sqrt{3}i}{2} \pm \sqrt{\frac{\sqrt{169+1200-13}}{2}}}$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{1369+13}{2}} - i\sqrt{\frac{1369-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{37+13}{2}} - i\sqrt{\frac{37-13}{2}} \right)$$

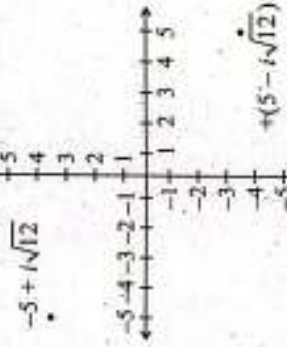
$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{50}{2}} - i\sqrt{\frac{24}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm (\sqrt{25} - i\sqrt{12})$$

$$\sqrt{13-20\sqrt{3}i} = \pm (5-2\sqrt{3}i)$$

Either

$$\sqrt{13-20\sqrt{3}i} = \pm (5-2\sqrt{3}i) \text{ or } \sqrt{13-20\sqrt{3}i} = \pm (-5+2\sqrt{3}i)$$



Q9. Find the value of x and y if

$$(-7 + i)(x + iy) + (-1 - 5i) = i(11 - i)$$

$$\text{Sol. } (-7 + i)(x + iy) - 1 - 5i = 11i - i^2$$

$$(-7 + i)(x + iy) - 1 - 5i = 11i - (-1)$$

$$(-7 + i)(x + iy) - 1 - 5i = 11i + 1$$

$$(-7 + i)(x + iy) = 11i + 1 + 1 + 5i$$

$$(-7 + i)(x + iy) = 2 + 16i$$

$$x + iy = \frac{2+16i}{-7+i}$$

$$x + iy = \frac{2+16i}{-7+i} \times \frac{-7-i}{-7-i}$$

$$x + iy = \frac{-14-2i-112i-16i^2}{49-(-1)}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{-14-2i-112i-16(-1)}{49+1}$$

$$x + iy = \frac{2-114i}{50}$$

$$x + iy = \frac{2}{50} - \frac{114}{50}i$$

$$x + iy = \frac{1}{25} - \frac{57}{25}i$$

By comparing the real and imaginary parts

$$x = \frac{1}{25} \text{ and } y = -\frac{57}{25}$$

Q10. Find the value of x and y if

$$(5-2i)(x+iy) + 3 = i(11-i) - 4i$$

$$\text{Sol. } (5-2i)(x+iy) + 3 = 11i - i^2 - 4i$$

$$(5-2i)(x+iy) + 3 = 11i - (-1) - 4i$$

$$(5-2i)(x+iy) = 11i + 1 - 4i - 3$$

$$(5-2i)(x+iy) = -2 + 7i$$

$$x + iy = \frac{-2+7i}{5-2i}$$

$$x + iy = \frac{-2+7i}{5-2i} \times \frac{5+2i}{5+2i}$$

$$x + iy = \frac{-10-4i+35i+14i^2}{25-4i^2}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25-4(-1)}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

$$x + iy = \frac{-10-4i+35i+14(-1)}{25+4}$$

Q12. If $z_1 = 4 + 5i$ and $z_2 = \alpha - 2i$, find the value of α such that $\text{Re}(z_1 z_2) = 20$.

Sol. For $z_1 z_2$

$$z_1 z_2 = (4 + 5i)(\alpha - 2i)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10i^2$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10(-1)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i + 10$$

$$z_1 z_2 = 4\alpha + 10 - 8i + 5\alpha i$$

$$z_1 z_2 = (4\alpha + 10) - (8 - 5\alpha)i$$

$$\text{Re}(z_1 z_2) = 4\alpha + 10 \text{ and } \text{Im}(z_1 z_2) = -(8 - 5\alpha)$$

$$\text{Given } \text{Re}(z_1 z_2) = 20$$

$$\text{Put the value of } \text{Re}(z_1 z_2)$$

$$4\alpha + 10 = 20$$

$$4\alpha = 20 - 10$$

$$4\alpha = 10$$

$$\alpha = \frac{10}{4} \Rightarrow \alpha = \frac{5}{2}$$